BASIC EXPLANATION GUIDE

Diploma in Cyber Security and Ethical Hacking with AI

Networking • Security • Ethical Hacking • Web Security • Defence • AI

PROGRAM OVERVIEW

About This Diploma

This diploma introduces the core skills needed to protect systems and networks, and to think like an attacker *legally* in order to find weaknesses before criminals do. It also shows how Artificial Intelligence (AI) can speed up security work such as reading logs and spotting threats.

#	Module	Key Topics
1	**Networking and Linux**	TCP/IP, DNS, HTTP, command line, permissions
2	**Security Fundamentals**	CIA triad, cryptography, IAM, security policies
3	**Ethical Hacking Basics**	Recon, scanning, enumeration, legal and ethical limits
4	**Web Application Security**	OWASP Top 10, injection, authentication flaws
5	**Defence and Monitoring**	Firewalls, SIEM basics, logs, incident response
6	**Labs, Reporting and AI**	CTF practice, vulnerability reports, AI for log analysis

Who is it for?

Beginners, students and IT professionals who want a career as a security analyst, penetration tester, SOC analyst or security engineer. No advanced prior knowledge is required — only curiosity and a strong sense of ethics.

Golden Rule

Only test systems you own or have **written permission** to test. Unauthorised hacking is illegal and unethical.

MODULE 1

Networking and Linux

Topics: TCP/IP, DNS, HTTP, command line, permissions

Every attack and every defence happens on a network, and most security tools run on Linux. This module builds that foundation.

TCP/IP

- The set of rules computers use to communicate. **IP** gives every device an address and routes data; **TCP** makes sure data arrives complete and in order.
- Key ideas: IP addresses, subnets, ports (e.g. 80 for HTTP, 443 for HTTPS, 22 for SSH) and the three-way handshake (SYN, SYN-ACK, ACK).

DNS (Domain Name System)

- Works like the internet's phonebook: it turns names like *example.com* into IP addresses.
- Security link: attackers abuse DNS through spoofing and poisoning, so defenders monitor it closely.

HTTP / HTTPS

- The protocol browsers use to talk to web servers using requests (GET, POST) and responses (status codes such as 200, 404, 500).
- HTTPS adds encryption (TLS) so data cannot be easily read in transit.

Linux Command Line

- Common commands: `ls, cd, pwd, cat, grep, ping, ip a, ssh`.
- The terminal is faster and more powerful than a graphical interface, and essential for security work.

File Permissions

- Each file has read (r), write (w) and execute (x) rights for the **owner, group and others**, e.g. `rwxr-xr--`.
- Change them with `chmod` and `chown`. Wrong permissions are a very common security weakness.

Remember: You cannot secure or test what you do not understand. Strong networking and Linux basics make every later module easier.

MODULE 2

Security Fundamentals

Topics: CIA triad, cryptography, IAM, security policies

These are the core principles behind almost every security decision.

The CIA Triad

- **Confidentiality** – only authorised people can see the data.
- **Integrity** – data is accurate and has not been changed without permission.
- **Availability** – systems and data are accessible when needed.

Cryptography

- **Symmetric encryption** uses one shared key (e.g. AES) — fast and used for bulk data.
- **Asymmetric encryption** uses a public and private key pair (e.g. RSA) — used for key exchange and digital signatures.
- **Hashing** (e.g. SHA-256) creates a fixed fingerprint of data to check integrity; passwords should be stored as salted hashes.

IAM – Identity and Access Management

- Controls *who* can access *what*. Covers authentication (proving identity), authorisation (what you may do) and accounting (logging actions).
- Best practices: strong passwords, multi-factor authentication (MFA), and **least privilege** — give only the access that is needed.

Security Policies

- Written rules that guide behaviour: password policy, acceptable use, data handling, incident response and backup policies.
- Policies, awareness training and technical controls work together.

Remember: Good security balances all three parts of the CIA triad. Protecting confidentiality at the cost of availability can also be a failure.

MODULE 3

Ethical Hacking Basics

Topics: Recon, scanning, enumeration, legal and ethical limits

Ethical hackers use the same techniques as attackers, but with permission, to find and fix weaknesses. A typical test follows these phases.

Reconnaissance (Recon)

- Gathering information about a target. **Passive** recon uses public sources (search engines, WHOIS, public records) without touching the target; **active** recon interacts with it directly.

Scanning

- Finding live hosts, open ports and running services. A common tool is **Nmap**.
- Scanning also checks for known vulnerabilities in the discovered services.

Enumeration

- Extracting more detail from discovered services: software versions, user accounts, shares and configurations, to understand where weaknesses may exist.

Legal and Ethical Limits

- Always get **written authorisation** and agree the scope (what may and may not be tested) before starting.
- Respect privacy, avoid damaging systems or data, and report findings responsibly to the owner.
- Laws such as India's IT Act, 2000 and similar laws worldwide make unauthorised access a crime.

Remember: Permission and scope are what separate an ethical hacker from a criminal.

MODULE 4

Web Application Security

Topics: OWASP Top 10, injection, authentication flaws

Websites and web apps are the most common targets. OWASP (Open Worldwide Application Security Project) publishes a Top 10 list of the most critical web risks.

OWASP Top 10 – examples of risk areas

- Broken access control (users reaching data they should not)
- Cryptographic failures (weak or missing encryption)
- Injection (e.g. SQL injection)
- Insecure design and security misconfiguration
- Vulnerable and outdated components
- Identification and authentication failures
- Logging and monitoring failures

Injection

- Happens when untrusted input is treated as part of a command or query, letting an attacker change what the application does (SQL injection, command injection, cross-site scripting).
- Prevention: **validate input**, use **parameterised queries**, and encode output.

Authentication Flaws

- Weak passwords, no rate-limiting on login, predictable or exposed session tokens, and missing MFA.
- Prevention: strong password rules, MFA, account lockout or throttling, secure session handling and proper logout.

Remember: Most web flaws come from trusting user input. Never trust data you did not create. Practise only on legal, intentionally vulnerable labs.

MODULE 5

Defence and Monitoring

Topics: Firewalls, SIEM basics, logs, incident response

Prevention is not enough. Defenders must also detect attacks quickly and respond effectively.

Firewalls

- Filter network traffic using rules: allow or block based on IP address, port and protocol.
- Types include network firewalls, host-based firewalls and web application firewalls (WAF).

Logs

- Records of events: logins, errors, file access and network connections. Logs are the main evidence for finding what happened.

SIEM (Security Information and Event Management)

- Collects logs from many sources, correlates them and raises alerts on suspicious patterns (e.g. many failed logins followed by a success).
- Used daily in a Security Operations Center (SOC). Examples: Splunk, Wazuh, Elastic Security.

Incident Response – main stages

- **1. Preparation 2. Detection and Analysis 3. Containment 4. Eradication 5. Recovery 6. Lessons Learned**

Remember: Assume that a breach can happen. A prepared team with good logs and a clear plan limits the damage.

MODULE 6

Labs, Reporting and AI

Topics: CTF practice, vulnerability reports, AI for log analysis

Skills are built by practice. This module connects hands-on learning with professional reporting and modern AI tools.

CTF (Capture The Flag) Practice

- Legal challenges where you find hidden “flags” by solving security puzzles in web, crypto, forensics and Linux categories.
- Safe platforms include TryHackMe, Hack The Box and OverTheWire.

Vulnerability Reports

- A good report is clear and actionable. Include: **summary, scope, steps to reproduce, evidence (screenshots), impact, severity rating and recommended fix**.
- Severity is often rated using CVSS (Common Vulnerability Scoring System).

AI for Log Analysis

- AI and machine learning can scan huge volumes of logs to spot unusual behaviour, group similar alerts and summarise incidents in plain language.
- Helpful uses: anomaly detection, alert triage, drafting report text and explaining unfamiliar log entries.
- Limits: AI can make mistakes or miss threats, so a human must verify results. Never paste sensitive or confidential data into public AI tools.

Remember: AI is an assistant, not a replacement. Combine AI speed with human judgement and ethics.

SUMMARY

Learning Path at a Glance

Start with **networking and Linux**, learn the **principles of security**, then practise **ethical hacking** and **web security** in legal labs. Learn **defence and monitoring** to see both sides, and finish by building **reporting skills** and using **AI** to work smarter.

Possible Career Roles

- Security / SOC Analyst
- Junior Penetration Tester
- Vulnerability Analyst
- Incident Responder
- Security Engineer

Note: This guide is a basic introduction for learning purposes. Always practise ethically and legally.